
Lab-to-Field Transfer in Musculoskeletal Reinforcement Learning: Comparing EMG/GRF-Informed and Markerless Reference Policies for Gait Simulation

Brock Perreca
Kahlert School of Computing
University of Utah
u1381052@utah.edu

Brian Keller
Department of Biomedical Engineering
University of Utah
Brian.keller@utah.edu

Abstract

Musculoskeletal reinforcement learning enables non-invasive estimation of internal joint forces and muscle activations, but the quality of reference motion data used during training remains a critical and underexplored bottleneck. We investigate how demonstration data quality affects imitation learning in a high-dimensional musculoskeletal control setting by training agents in MyoSuite—a physiologically accurate simulation environment with 80 muscle actuators and 20 degrees of freedom—on two matched reference motion sources: rich laboratory data (marker-based motion capture with force plates and EMG) and low-cost markerless smartphone video processed through OpenCap. We hypothesize that agents trained on less rich kinematic data will produce significantly less biomechanically valid emergent behavior, specifically less realistic ground reaction forces and muscle activation patterns, even when those quantities are never directly included in the reward signal. We propose six experimental conditions spanning hand-crafted reward ablations and Generative Adversarial Imitation Learning (GAIL), evaluated against ground-truth force plate and EMG recordings, to provide a direct empirical answer to the clinically motivated question: how much biomechanical fidelity is lost when a \$150,000 motion capture lab is replaced with two iPhones?

1 Introduction

Understanding human movement at the level of individual muscle forces and joint contact loads is central to clinical biomechanics, rehabilitation, and assistive device design. Traditionally, estimating these internal quantities requires expensive instrumented laboratories equipped with optical marker-based motion capture systems, force plates, and electromyography (EMG) sensors—infrastructure that costs upwards of \$150,000 and requires trained operators. Musculoskeletal reinforcement learning offers a compelling alternative: by training a simulated agent with physiologically accurate muscles to imitate recorded human motion, internal forces and activations emerge as a byproduct of the physics, without the need for invasive measurement [9, 2].

However, all existing musculoskeletal RL work relies on clean, laboratory-grade reference motions. Recent advances in markerless motion capture—particularly OpenCap [10], which reconstructs 3D kinematics from two smartphone cameras—promise to democratize motion data collection at a fraction of the cost. Yet OpenCap introduces known kinematic noise, particularly in the frontal and transverse planes [6], and no study has directly measured what this noise costs a musculoskeletal RL agent in terms of emergent biomechanical validity.

This project fills that gap. Using the OpenCap SimTK validation dataset—which provides simultaneous marker-based and markerless recordings of the same subjects and trials alongside ground-truth force plate and EMG data—we train matched RL agents under progressively richer reference conditions and measure how emergent ground reaction forces (GRF), muscle activations, and joint contact forces change. We further test whether learned reward functions via GAIL [5] are more robust to reference noise than hand-crafted alternatives. This investigation directly engages core course themes of reward design, inverse reinforcement learning, and learning from demonstrations of varying quality.

2 Related Work

Physics-based motion imitation. DeepMimic [8] established the modern framework for training physically simulated characters to imitate motion capture data using deep RL, introducing Reference State Initialization (RSI) and a weighted composite reward over pose, velocity, end-effector, and root terms. This reward formulation remains the standard baseline for hand-crafted imitation rewards and is the foundation for conditions R1–R4 in our experimental design.

Musculoskeletal RL. KINESIS [9] extends the DeepMimic framework to a physiologically accurate MyoSuite [1] lower-body model with 80 muscle actuators and 20 degrees of freedom, training on 1.9 hours of KIT motion capture data. Crucially, KINESIS demonstrates that emergent muscle activations correlate with recorded EMG without explicit EMG reward terms—establishing the physiological plausibility bar against which our conditions are measured. KinTwin [2] scales this approach to whole-body biomechanical models trained on 34 hours of data from 467 participants, including individuals with movement impairments, and validates emergent GRFs against instrumented walkway measurements. Both systems assume clean laboratory-grade reference motion; neither has been tested with markerless input.

Markerless motion capture. OpenCap [10] reconstructs 3D joint kinematics from two smartphone cameras, achieving approximately 4.5 degree joint angle RMSE for walking and knee adduction moment $r^2 = 0.80$ versus marker-based pipelines with force plates. Horsak et al. [6] further characterize OpenCap’s accuracy across physiological and pathological gait patterns, reporting RMSE of 3.7–10.2 degrees with degraded performance in frontal and transverse planes. These studies establish the input noise profile our agents must contend with, but neither connects markerless error to downstream RL policy quality.

Adversarial imitation learning. GAIL [5] frames imitation as occupancy measure matching via a GAN-style discriminator, bypassing explicit reward engineering. We use GAIL to test whether learned rewards are more robust to noisy reference data than hand-crafted alternatives—a comparison that, to our knowledge, has not been made in the musculoskeletal domain.

Kinematics-to-forces prediction. Liew et al. [7] train deep neural networks to predict joint contact forces from joint angles alone, achieving errors within minimal detectable change values. We adopt this approach as a post-hoc evaluation tool: rollout joint angles from each condition are passed through a Liew-style model to estimate joint contact forces, enabling comparison across conditions without requiring force data during training.

Our work differs from all of the above by directly and systematically comparing RL policy quality across matched lab-grade and markerless reference motions from the same subjects and trials, providing the first controlled measurement of what markerless noise costs in emergent biomechanical validity.

3 Method

3.1 Environment and Model

We use the MyoSuite lower-body musculoskeletal model: MyoLegs with 80 Hill-type muscle actuators and 20 skeletal degrees of freedom, simulated in MuJoCo. Training proceeds in two stages. In Stage 1, we train a base walking policy from scratch using Soft Actor-Critic (SAC) [4] on the native myoLegWalk-v0 task reward, which combines forward velocity, cyclic hip, and upright torso terms. SAC’s off-policy replay buffer and automatic entropy tuning make it substantially more sample-efficient than on-policy alternatives for this setting, where the action space is 80-dimensional

and activations are bounded to $[0, 1]$. The actor uses a Beta-distribution parameterization over muscle activations, which is more natural than Gaussian for actions constrained to the unit interval. This stage produces a locomotion prior that walks stably without any reference motion data.

In Stage 2, a GAIL discriminator is fine-tuned from this base policy. Rather than operating in the full 403-dimensional MyoSuite observation space, the discriminator works in a reduced 48-dimensional space: a 32-dimensional IK observation (joint angles and velocities for pelvis, hips, knees, ankles, and lumbar) concatenated with the 16 EMG-relevant muscle activations sliced from the 80-dimensional action, keeping the discriminator input aligned with what expert data actually provides. The composite reward used during SAC-GAIL fine-tuning is

$$r = r_{\text{env}} + w_{\text{gail}} \cdot r_{\text{gail}} + w_{\text{sym}} \cdot r_{\text{sym}}, \quad (1)$$

where $r_{\text{gail}} = \log \sigma(D(\mathbf{s}_{\text{ik}}, \mathbf{a}_{\text{ik}}))$ is the standard GAIL reward [5], and $r_{\text{sym}} = -\|\boldsymbol{\theta}_R - \boldsymbol{\theta}_L\|^2$ penalizes squared differences between corresponding right and left leg joint angles, encouraging bilateral symmetry without requiring it to appear in the reference data. The discriminator is trained with binary cross-entropy, label smoothing (expert targets 0.9, policy targets 0.1 to prevent saturation), and a WGAN-GP gradient penalty [3] for training stability. Warm-starting GAIL from a walking prior avoids the cold-start failure mode in which the discriminator trivially distinguishes a non-locomoting policy from reference trajectories before any meaningful imitation gradient can form. All conditions use Reference State Initialization and early termination on fall detection.

3.2 Data

We use the OpenCap SimTK validation dataset: 10 subjects performing natural walking with simultaneous marker-based motion capture, dual-iPhone OpenCap recording, force plates, and EMG. From the same physical recordings, we derive two reference motion sets: (1) **Rich**: marker-based inverse kinematics + GRF-driven inverse dynamics + raw EMG activations, and (2) **Markerless**: OpenCap inverse kinematics output only.

3.3 Experimental Conditions

We define six conditions that systematically vary the reference data source and reward signal, as summarized in Table 1.

Table 1: Experimental conditions. R1–R4 use hand-crafted composite rewards with SAC; R5–R6 use SAC-GAIL fine-tuned from the same base walking policy, with the only difference being the reference trajectory provided to the discriminator.

ID	Reference Source	Reward Components	Purpose
R1	OpenCap IK only	Joint angle tracking	Markerless baseline
R2	Marker-based IK	Joint angle tracking	Clean kinematic baseline
R3	Marker IK + GRF/ID	Angles + joint moments	Kinetics enrichment
R4	Marker IK + GRF + EMG	Angles + moments + EMG	Full lab upper bound
R5	GAIL (OpenCap traj.)	Learned discriminator	Learned reward, markerless
R6	GAIL (lab traj.)	Learned discriminator	Learned reward, lab data

3.4 Reward Formulation

For R1–R4, we use a weighted composite reward following DeepMimic [8]: $r = w_p r_{\text{pose}} + w_v r_{\text{vel}} + w_r r_{\text{root}} + w_m r_{\text{moment}} + w_e r_{\text{emg}}$, where r_{pose} penalizes joint angle deviation from the reference, r_{vel} tracks joint velocities, r_{root} tracks pelvis position and orientation, r_{moment} matches inverse-dynamics joint moments (R3 and R4 only), and r_{emg} computes cosine similarity to normalized EMG recordings (R4 only). For R5–R6, the reward is the SAC-GAIL composite described in Section ??: $r = r_{\text{env}} + w_{\text{gail}} \cdot r_{\text{gail}} + w_{\text{sym}} \cdot r_{\text{sym}}$, where the discriminator D operates on the projected IK subspace $(\mathbf{s}_{\text{ik}}, \mathbf{a}_{\text{ik}})$ rather than the full observation, and the symmetry term r_{sym} is held constant across R5 and R6.

4 Experimental Design

4.1 Hypotheses

H1: Agents trained on richer reference data produce more biomechanically valid emergent behavior: $R4 > R3 > R2$ on GRF correlation and EMG Pearson r , even though GRF and EMG are not part of the reward for R2.

H2: OpenCap kinematic noise measurably degrades emergent biomechanical fidelity: R1 performs worse than R2 on GRF and EMG metrics despite identical reward structure, with the largest degradation in frontal and transverse plane variables.

H3: GAIL’s learned reward is more robust to noisy references than hand-crafted rewards: the gap between R5 and R6 is smaller than the gap between R1 and R2.

4.2 Evaluation Metrics

Primary (biomechanical validity): (1) GRF shape correlation between emergent and measured force plate data—agents in R1, R2, and R5 never see GRF during training; (2) per-muscle EMG Pearson r between emergent activations and recorded EMG; (3) joint contact force (JCF) estimates obtained by passing rollout joint angles through a Liew et al. [7] deep neural network, compared across conditions.

Secondary (policy quality): kinematic RMSE versus reference joint angles, frame coverage (percentage of reference trajectory successfully tracked), gait stability (episode length, fall rate), and spatiotemporal parameters (step length, cadence, stance/swing ratio).

4.3 Key Comparisons

R1 vs. R2 isolates the OpenCap domain gap at equal reward complexity. R2 vs. R3 vs. R4 provides a kinetic enrichment ablation—quantifying how much force and EMG reward terms add beyond kinematics-only tracking. R1 vs. R5 and R2 vs. R6 test whether GAIL’s learned reward generalizes better than hand-crafted rewards under noisy and clean references, respectively.

4.4 Training Infrastructure

All conditions will be trained on an NVIDIA RTX 5090. Stage 1 SAC pretraining uses a single MuJoCo environment with a replay buffer of 10^6 transitions; Stage 2 SAC-GAIL fine-tuning similarly operates on-policy collection with a separate discriminator replay. Estimated wall-clock time is 12–24 hours per condition. Training curves and reward component breakdowns will be logged via Weights & Biases for interpretability and reproducibility.

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